Calcurion Maths Worksheets

A-Level Further Mathematics

Topic: Further Pure 1 – Inequalities

Medium & Hard • 25 Questions

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Section A – Medium (Q1–15)

1. Solve $x^2 - 5x + 6 > 0$. **[3]**

2. Solve $\frac{x-1}{x+2} > 0$. **[4]**

3. Solve $|x - 3| < 5$. **[3]**

4. Solve $|2x + 1| \geq 3$. **[3]**

5. Solve $\frac{2x-1}{x-3} \leq 1$. **[4]**

6. Solve $x^2 + 2x - 8 \leq 0$. **[3]**

7. Solve $|x + 1| + |x - 2| > 5$. **[5]**

8. Solve $\frac{x}{x-1} > 2$. **[4]**

9. Solve $x^3 - x > 0$. **[4]**

10. Solve $\frac{1}{x} < \frac{1}{x+1}$. **[4]**

11. Find the set of values of x for which $x^2 - 4x + 3 > 0$ and $x > 2$. **[3]**

12. Solve $|x^2 - 4| < 3$. **[4]**

13. Solve $\frac{x^2-1}{x+2} \geq 0$. **[4]**

14. Solve $2^x > 4^{x-1}$. **[4]**

15. Solve $\ln(x - 1) > \ln(3 - x)$. **[4]**

Section B – Hard (Q16–25)

16. Solve $\frac{x+1}{x-2} - \frac{2}{x+1} > 1$. **[6]**

17. Solve $|x - 1| + |x + 2| + |x - 3| \leq 10$. **[5]**

18. Find the range of values of k for which $x^2 + kx + 4 > 0$ for all x. **[4]**

19. Solve $\frac{x^2-5x+6}{x^2-4} \geq 0$. **[5]**

20. Solve $e^{2x} - 3e^x + 2 > 0$. **[5]**

21. Solve $\frac{|x|-1}{|x|+2} < 1$. **[4]**

22. Find x such that $\frac{2x+1}{x-1} > \frac{x+2}{x+1}$. **[5]**

23. Solve $x^4 - 5x^2 + 4 \leq 0$. **[5]**

24. Solve $\sqrt{x+2} > x$. **[5]**

25. Solve $\frac{1}{x^2-1} < \frac{1}{x-1}$. **[5]**

End of Worksheet • *Show all working clearly.*